

Storm Sprinklers CRM

Comprehensive Pre-Launch Test Plan

Version: 1.0

Date: July 2026

Purpose: Verify every CRM feature before active team use

Environment: Staging / Production (check each box for the environment tested)

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How to Use This Document

This plan contains **262** manual test cases across **21 modules** covering every feature of the Storm Sprinklers CRM.

Instructions

- Work through sections **in order** for first pass, or jump to specific modules for targeted retesting.
- Each row has a unique **ID** (e.g. AUTH-01) for referencing bugs and retests.
- Mark **Pass** column with ☒ when the expected result is confirmed.
- Use the **Notes** column for bug IDs, screenshots, or environment details.
- Section **20 (E2E)** should be run last after individual modules pass.
- Section **21 (Pre-Go-Live)** is the final gate before handing to your team.

Recommended test order

1. Settings & configuration (Section 18) — set up the system first
2. Price Book & Checklists (Sections 8, 10) — catalog and field workflows
3. Customers & Schedule (Sections 3, 4) — core data entry
4. Visits & field workflow (Section 5) — the daily technician experience
5. Inbox & Voice (Sections 6, 7) — communication channels
6. Estimates & Payments (Sections 9, 17) — sales and money
7. Maintenance Plans (Section 11) — recurring revenue
8. Portal & Booking (Sections 15, 16) — customer self-service
9. Marketing & Reporting (Sections 13, 14) — growth and analytics
10. E2E scenarios & Go-Live (Sections 20, 21) — final validation

Test accounts needed

ROLE	PURPOSE	CREATED?
ADMIN	Full access, settings, integrations, delete, refunds	☐
MANAGER	Schedule, employees, customers, most settings	☐
CSR	Inbox, scheduling, customer management, voice desk	☐
TECH	Field service — assigned visits, time tracking, checklists	☐
INSTALLER	Install crew member — crew visits only	☐
SALES	Leads, estimates, pipeline	☐
CUSTOMER	Portal login, booking, estimate approval, invoice pay	☐

External services to verify before testing

- **Twilio** — SMS, voice, MMS, email (phone number, webhooks registered)
- **Stripe** — Invoice pay, visit checkout, deposits, card-on-file, Connect payouts
- **Google** — Business Profile OAuth, Ads OAuth, Search Console OAuth
- **Meta** — Page token, webhooks for DMs and social insights
- **Slack** — Bot token, GBP review channel configured
- **Google Maps** — Address autocomplete, map embeds, driving ETA

- **Rachio** — API key, controller linked to test property
- **SerpAPI** — Keyword ranking maps (optional; mock data without key)
- **OpenAI** — AI descriptions, email generation, SEO recommendations (optional)
- **Blob storage** — Attachments, photos, voice clips, email logos

1. Authentication, Roles & Global Navigation

Verify login, role-based access, top navigation, and global shortcuts before testing feature modules. Test with at least one account per role: Admin, Manager, CSR, Tech, Installer, Sales.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
auth-01	Auth	Staff login with valid credentials	Go to /login. Enter active employee email and password. Submit.	Redirect to /home. Session shows correct name and role. Top nav visible.	☐	
auth-02	Auth	Staff login rejects invalid credentials	Enter wrong password for a valid user. Submit.	Error shown. No session created.	☐	
auth-03	Auth	Archived employee cannot log in	Archive an employee in Settings → Employees. Attempt login as that user.	Login fails. User cannot access CRM.	☐	
auth-04	Navigation	Primary navigation links work	Click each top nav item: Home, Customers, Inbox, Schedule, Timesheets, Price Book, Maintenance Plans, Marketing, Reporting, Settings.	Each module loads without error. Active state highlights correctly.	☐	
auth-05	Navigation	Mobile navigation menu	Resize to mobile width. Open hamburger menu. Navigate to 3 modules.	Menu opens/closes. Links work. Overlay dismisses on selection.	☐	
auth-06	New menu	New menu — create customer	Click + New → New customer. Fill name, phone, email. Save.	Customer created. Redirect or confirmation. Customer appears in list.	☐	
auth-07	New menu	New menu — create visit	Click + New → New visit. Fill title, division, date/time, technician, service area. Save.	Visit created on schedule. Opens visit detail or shows on board.	☐	
auth-08	New menu	New menu — create estimate	Click + New → New estimate. Select customer. Save.	Estimate created in DRAFT. Opens estimate builder.	☐	
auth-09	New menu	New visit blocks Do Not Service customer	Flag a customer Do Not Service. Try to create visit for them via New menu.	Booking blocked with clear error. No visit created.	☐	
auth-10	Notifications	Notification bell	Trigger a test notification (e.g. website lead). Open notification bell.	Unread badge increments. Item appears. Click marks read and navigates if linked.	☐	
auth-11	Voice	Phone dialer opens from top nav	Click phone icon in top nav.	Voice dialer dialog opens. Softphone status shown (ready/connecting/error).	☐	
auth-12	Roles	Field role cannot access office-only settings writes	Log in as TECH. Open Settings → Company. Attempt to save changes (or verify API 403).	View may work; save blocked for unauthorized roles.	☐	

2. Home Dashboard

The home page is the daily starting point for office and field staff.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
home-01	Home	Home summary cards load	Open /home as Admin.	Greeting shows first name. Estimates, Jobs, Invoices, Maintenance cards show counts and link correctly.	☐	
home-02	Time clock	Clock in	Click Clock In on home page.	Status shows clocked in with live timer. POST /api/time-clock succeeds.	☐	
home-03	Time clock	Clock out	While clocked in, click Clock Out.	Timer stops. Today's hours displayed. Entry appears on /timesheets.	☐	

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
home-04	Time clock	Double clock-in prevented	Clock in twice without clocking out.	Second attempt shows error or is disabled.	☐	
home-05	Home	KPI strip date range	Change KPI strip range: YTD, MTD, Last 30 days.	Metrics update. Link to /reporting works.	☐	

3. Customers Module

Covers customer list, profile, properties, irrigation, Rachio, leads, jobs, estimates, and invoices sub-pages.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
customers-01	List	Customer list search and filters	Go to /customers. Search by name. Filter by city, ZIP, lead source. Toggle Active/Archived.	Results filter correctly. Count updates. Clear filters resets view.	☐	
customers-02	List	Create customer from list	Click Create customer. Enter name, phone, email. Submit.	Customer created. Redirect to profile. Appears in active list.	☐	
customers-03	List	Bulk archive customers	Select 2+ customers. Bulk action → Archive. Confirm.	Customers move to Archived tab. Restore works from Archived tab.	☐	
customers-04	List	Bulk merge customers	Select 2 customers as Admin/Manager. Merge into target.	Records merged. Secondary customer data consolidated.	☐	
customers-05	List	Bulk Do Not Service flag	Select customer. Mark Do Not Service.	Badge appears on customer. Scheduling blocked elsewhere.	☐	
customers-06	Profile	Customer profile — edit contact	Open customer. Edit name, address (autocomplete), phone, email, lead source. Save.	Changes persist. Map embed updates if address changed.	☐	
customers-07	Profile	Alternate phone numbers	In edit mode, add alternate phone with note. Save. Delete one.	Phones listed with call/SMS actions. CRUD works.	☐	
customers-08	Profile	Customer tags	Add tag. Remove tag. Save as Admin/Manager.	Tags display on profile. Persist after reload.	☐	
customers-09	Profile	Send portal link	Click Send portal link for customer with email (portal enabled).	Email sent or success toast. Customer can use magic link.	☐	
customers-10	Profile	Customer notes and attachments	Add text note. Upload JPEG and PDF attachment. Delete attachment.	Note shows author/timestamp. Files upload, preview, and delete.	☐	
customers-11	Payments	Payment methods on file	As CSR/Manager, add card via Stripe Elements. Copy secure link. Email secure link.	Card saved in Stripe. Links work for customer self-service.	☐	
customers-12	Properties	Add property	Properties tab → Add property with name and address (autocomplete).	Property appears. Lat/Lng captured. Map shows location.	☐	
customers-13	Properties	Delete property	Delete a property with no critical dependencies.	Property removed from list.	☐	
customers-14	Irrigation	Irrigation wizard — full publish	Run 6-step wizard: aerial capture, system overview, zone map, conditions, irrigation, review. Publish.	Draft saves at each step. Publish sets status Published. Portal shows guide.	☐	
customers-15	Irrigation	Irrigation wizard — import from design	On a property linked to design project, import from design in wizard.	Zone data seeds correctly into wizard.	☐	
customers-16	Irrigation	Legacy irrigation editor	Set diagram URL and per-station run minutes. Save irrigation guide.	Data persists. Displays alongside wizard map if both exist.	☐	

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
customers-17	Rachio	Rachio link and zone run	Link Rachio device to property. View zones. Run zone for 5 minutes. Stop all.	Device shows online. Zone run starts. Stop all works.	☐	
customers-18	Visits	Create visit from customer	Visits tab → Add visit. Fill required fields. Save.	Visit created. Redirect to visit detail. Appears on schedule.	☐	
customers-19	Estimates	Create estimate from customer	Estimates tab → Create estimate.	DRAFT estimate created. Opens builder.	☐	
customers-20	Maintenance	Enroll in maintenance plan	Maintenance tab → Enroll in plan. Select template, property, billing frequency.	DRAFT enrollment created. Appears in enrollment list.	☐	
customers-21	Referrals	Referral enrollment	Enroll customer in referral program. Copy share link.	Member created. Share URL works for public referral form.	☐	
customers-22	Jobs	Jobs list	Go to /customers/jobs. Search. Click a job.	Jobs listed with status, date, total. Links to visit detail.	☐	
customers-23	Leads	Leads list and convert	Go to /customers/leads. Create lead. Convert to customer.	Lead appears. Convert creates customer. Status WON. View customer link works.	☐	
customers-24	Invoices	Invoices list actions	Go to /customers/invoices. Send, Remind, Copy pay link, Refund (Admin).	Each action succeeds or shows appropriate error. Pay link opens /pay/{token}.	☐	
customers-25	Leads inbox	Website leads inbox	Submit website form. Open /inbox/leads.	Submission appears within 15s. Detail shows form content. Delete works for email leads.	☐	

4. Schedule & Team Management

Covers schedule board, quick-add, needs-scheduling queue, and team schedules/time-off.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
schedule-01	Board	Schedule week view	Open /schedule. View week. Navigate prev/next/today.	Jobs display in correct time slots. Colors apply per color mode.	☐	
schedule-02	Board	Schedule day and month views	Switch to Day view, then Month view. Click a day in month view.	Day shows per-employee columns. Month shows counts. Click switches to day.	☐	
schedule-03	Filters	Filter by service area and employee	Toggle areas and hide/show employees in sidebar.	Grid filters correctly. Unassigned column toggles.	☐	
schedule-04	Quick-add	Quick-add visit from slot	Click empty time slot. Fill customer, technician, division. Save.	Visit appears on board at correct time. Default 3-hour window applied.	☐	
schedule-05	Metrics	Schedule revenue summary	View week with completed/scheduled jobs.	Daily/weekly revenue and hours metrics display.	☐	
schedule-06	Needs scheduling	Needs scheduling queue	Approve design estimate with deposit. Open /schedule/needs-scheduling.	Estimate appears in queue with customer, total, duration. Links work.	☐	
schedule-07	Team	Team schedule — set work days	Team schedules → select employee. Set Mon–Fri 8am–5pm working. Save.	Schedule persists. Non-working days block assignment server-side.	☐	
schedule-08	Time off	Request time off (Tech)	Log in as Tech. Submit PTO request for future dates.	Request shows as PENDING. Appears in manager pending queue.	☐	
schedule-09	Time off	Approve time off (Manager)	As Manager, approve pending time off request.	Status APPROVED. Employee unavailable on those dates for scheduling.	☐	

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
schedule-10	Time off	Assignment blocked on approved time off	Try scheduling visit for employee on approved time-off date.	Server returns error. Assignment blocked.	☐	
schedule-11	Validation	Double-booking prevention	Assign overlapping visits to same technician.	Second assignment blocked with conflict error.	☐	

5. Visit Detail & Field Workflow

Complete visit lifecycle from scheduled through payment.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
visits-01	Schedule	Edit visit schedule	As Manager, change date, time, assigned technician. Save.	Schedule updates. Customer notification sent if configured.	☐	
visits-02	Roles	Tech cannot edit schedule	Log in as Tech on assigned visit. Verify schedule fields read-only.	No edit controls for date/technician.	☐	
visits-03	Time tracking	On my way with ETA	Start On my way. Allow geolocation. Verify ETA banner.	Status EN_ROUTE. ETA shown. Customer SMS sent if configured.	☐	
visits-04	Time tracking	Start, pause, resume work	Start my time → Pause → Resume.	Status transitions IN_PROGRESS/PAUSED. Timer runs correctly.	☐	
visits-05	Checklists	Finish visit — required checklists	Leave required checklist incomplete. Try Finish visit.	Blocked with message listing incomplete checklists.	☐	
visits-06	Checklists	Complete all checklist item types	Fill Pass/Flag/Fail, Number, Note, Photo, Multi-select, Select one, Checkbox items. Mark complete.	Each type saves. Checklist marked COMPLETED.	☐	
visits-07	Callback	Callback flag	Toggle Callback job checkbox. Verify callback-specific checklists excluded.	Flag persists. Matching rules respect excludeCallbacks.	☐	
visits-08	Line items	Add line items from price book	Add service and material from ItemPicker. Adjust quantity.	Line items appear. Totals recalculate. Subtotal/discount/total correct.	☐	
visits-09	Discounts	Add visit discount	Add FIXED and PERCENT discounts.	Totals update correctly. PERCENT applies to subtotal.	☐	
visits-10	Notes	Visit notes and attachments	Add note. Upload photo and PDF.	Note shows author/time. Attachments viewable and deletable.	☐	
visits-11	Parts run	Parts run	Open Parts run. View nearest suppliers. Navigate (pauses timer if in progress).	Suppliers listed. Google Maps opens. Timer pauses if applicable.	☐	
visits-12	Payments	Collect payment via Stripe	Click Collect payment on visit with balance > 0.	Stripe Checkout opens. On success, visit shows Paid. Invoice synced.	☐	
visits-13	Profit	Profit section (Admin only)	View visit as Admin. Verify profit section visible.	Margin data loads. Hidden for Tech role.	☐	
visits-14	Delete	Delete visit	As Admin, delete visit with confirmation.	Visit removed. No longer on schedule.	☐	
visits-15	Install	Install plan section	On install visit with design, set install days. Save.	Install days persist. Design zone viewer loads.	☐	
visits-16	Maintenance	Link maintenance plan visit	On visit with enrolled customer, assign plan visit from dropdown.	Plan visit linked. Plan discounts applied to line items.	☐	
visits-17	Irrigation	Irrigation map editor on visit	Edit map: place markers, draw zones, set zone attributes. Save draft.	Map saves. Status shows Draft. Published map visible in portal.	☐	

6. Inbox — Voice, SMS, Email, Social, Leads

Test all communication channels and CSR workflows.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
inbox-01	CSR Desk	CSR Desk — live queue	Place inbound test call. View CSR Desk queue.	Caller appears in queue with lookup info (name, city, last visit).	☐	
inbox-02	Voice	Accept queued call	On Voice → Customers, click Accept on queue entry.	Call connects to softphone. Active call bar appears.	☐	
inbox-03	Voice	Outbound dial — manual number	Open dialer. Enter phone. Place call.	Call connects. History logs outbound entry.	☐	
inbox-04	Voice	Outbound dial — customer search	Dialer → Find customer. Search and select. Call.	Customer phone populated. Call placed.	☐	
inbox-05	CSR Desk	Book appointment during call	During active call with known customer, Book appointment. Fill fields. Save.	Visit created. Banner confirms. Disposition links on wrap-up.	☐	
inbox-06	Voice	Call wrap-up disposition	End call. Select disposition (Booked / Not booked / Non-opportunity). Add note. Save.	Disposition saved. Appears in call history.	☐	
inbox-07	Voice	Warm transfer	During call, Transfer → Warm → select available agent. Complete transfer.	Consult connects. Complete transfer assigns new handler.	☐	
inbox-08	Voice	Cold transfer	Transfer → Cold → select agent.	Customer redirected. CSR disconnected.	☐	
inbox-09	Voice	Hold and mute	During call, toggle Mute and Hold.	Mute silences mic. Hold plays hold music to customer.	☐	
inbox-10	Voice	Call recording playback	Complete a recorded call. Open call detail. Play recording.	Recording plays. Transcript shown if enabled.	☐	
inbox-11	SMS	SMS — send to customer	SMS → Customers. Select thread or compose. Send text message.	Message delivered. Appears in thread. Customer receives SMS.	☐	
inbox-12	SMS	SMS — send MMS attachment	Attach image to SMS. Send.	MMS delivered. Media displays in thread.	☐	
inbox-13	SMS	SMS — block contact	Block a customer contact. Attempt to send SMS.	Send blocked with 403. Blocklist enforced.	☐	
inbox-14	SMS	SMS — contact info detection	Receive inbound SMS with name/email. Apply contact info dialog.	Fields parsed. Customer created/updated. Conversation linked.	☐	
inbox-15	SMS	SMS — team internal	SMS → Team. Message another employee.	Internal thread created. Message delivered.	☐	
inbox-16	Email	Email compose and send	Go to /inbox/compose. Add recipient, subject, body, attachment. Send.	Email sent. Appears in sent folder (API).	☐	
inbox-17	Social	Facebook DM reply	Open Facebook DMs thread. Reply with text.	Message sent via Meta API. Appears in thread.	☐	
inbox-18	Social	Instagram DM reply	Open Instagram DMs thread. Reply.	Message sent. Within 24-hour window enforced.	☐	
inbox-19	Voice	Team voice directory	Voice → Team. Click-to-call team member.	Internal call placed via softphone.	☐	

7. Voice Settings

Configure phone system before go-live.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
voice-settings-01	Numbers	Add phone number manually	Settings → Voice → Numbers. Add E.164 number with friendly name.	Number appears in list.	☐	
voice-settings-02	Numbers	Import numbers from Twilio	Click Import from Twilio.	Twilio numbers sync to list.	☐	
voice-settings-03	Numbers	Buy Twilio number	Search by area code. Purchase number.	Number purchased and added to account.	☐	
voice-settings-04	Numbers	Set primary number	Mark a number as primary.	Primary badge shown. Used as default caller ID.	☐	
voice-settings-05	Flows	Create call flow with IVR	Settings → Voice → Flows. Create flow with IVR node (digits → branches).	Flow saves. Inbound calls route per digit selection.	☐	
voice-settings-06	Groups	Create agent group	Settings → Voice → Groups. Create group with ring strategy and members.	Group saves. DIAL_GROUP node rings members per strategy.	☐	
voice-settings-07	Clips	Upload audio clip	Settings → Voice → Clips. Upload MP3/WAV.	Clip available for IVR prompts.	☐	
voice-settings-08	Hours	Business hours	Settings → Voice → Hours. Set Mon–Fri open, Sat–Sun closed. Save.	After-hours routing uses afterHoursNodeId on inbound calls.	☐	
voice-settings-09	Overview	Record and transcribe toggles	Settings → Voice. Toggle Record calls and Transcribe calls.	New calls record/transcribe per settings.	☐	
voice-settings-10	Inbox settings	Register Twilio webhooks	Settings → Inbox → Register Twilio webhooks.	Webhooks registered. SMS and voice inbound work.	☐	

8. Price Book

Catalog management for services, materials, pricing forms, templates, and discounts.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
pricebook-01	Services	Create service category and item	Price Book → Create category. Add service with name, price, description.	Category and item appear. Searchable from ItemPicker.	☐	
pricebook-02	Services	Flat-rate calculated service	Enable flat rate in settings. Create service with labor rate + hours + bundled materials.	Price auto-calculates. Breakdown shown. Manual override option works.	☐	
pricebook-03	Materials	Material with markup tiers	Enable material markups. Create material with cost. Enable markup on item.	Sell price = cost × (1 + tier %). Tier matched by cost range.	☐	
pricebook-04	Import	CSV import services	Export CSV. Modify. Re-import with update existing toggle.	Items created/updated. Toast shows counts.	☐	
pricebook-05	Labor rates	Labor rates CRUD	Settings → Price Book → Labor rates. Create rate. Set as default.	Rate saves. Calculated services recalculate on default change.	☐	
pricebook-06	Markups	Material markup tiers	Edit tier rows. Save.	Materials with markupEnabled recalculate.	☐	
pricebook-07	Bulk adjust	Bulk price adjust	Bulk adjust +10% on all services. Preview then Apply.	Prices update. Calculated services recalculate.	☐	
pricebook-08	AI	AI description generation	On service item, click Generate description (requires OpenAI key).	Description populated by AI.	☐	

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
pricebook-09	Pricing forms	Create pricing form	Pricing forms → Create form by name.	Form created with default fields.	☐	
pricebook-10	Templates	Create estimate template	Estimate Templates → Create with name and line item.	Template appears in list.	☐	
pricebook-11	Discounts	Create catalog discount	Discounts → Create PERCENT and FIXED discounts. Toggle active.	Discounts listed with status.	☐	

9. Estimates

Estimate builder, send, approve, convert, and portal flow.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
estimates-01	Builder	Build estimate with line items	Open estimate. Add items from price book. Add discount. Verify totals.	Subtotal, discount, total correct. No tax line.	☐	
estimates-02	Builder	Add estimate note and attachment	Add text note. Upload image attachment.	Note and attachment appear.	☐	
estimates-03	Send	Send estimate to customer	Click Send. Verify customer has email or phone.	Status SENT. Notification sent. Expiry date set.	☐	
estimates-04	Signature	Staff signature approval	Draw signature on canvas. Save.	Status APPROVED. signedAt recorded.	☐	
estimates-05	Convert	Copy estimate to visit	With linked visit, Copy to visit. Select this visit or new visit.	Line items and discounts copied. Status CONVERTED.	☐	
estimates-06	Portal	Portal estimate approval	Open /portal/{slug}/estimates/{token} as customer. Sign and approve.	Status APPROVED. Deposit checkout if required.	☐	
estimates-07	Portal	Portal premium tier selection	On design estimate with premium option, select Premium tier. Approve.	Total updates to premiumOptionTotal.	☐	
estimates-08	Payments	Deposit payment flow	Approve estimate requiring deposit. Complete Stripe checkout.	Deposit paid. Install visit auto-created if configured.	☐	
estimates-09	Expiry	Estimate expiry	Send estimate. Wait past expiry or backdate expiresAt. Reload.	Status auto-flips to EXPIRED.	☐	
estimates-10	Delete	Delete estimate	As Admin/Manager, delete DRAFT estimate.	Estimate removed.	☐	
estimates-11	Design	Design estimate — needs scheduling	Approve design estimate with deposit. Check needs-scheduling queue.	Estimate appears. Install visit created.	☐	

10. Checklists

Template configuration and visit execution.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
checklists-01	Templates	Create checklist template	Settings → Checklists → New. Name, divisions, items of each type. Save.	Template appears in list with rule summary.	☐	
checklists-02	Matching	Apply to all jobs rule	Create template with Apply to all jobs. Schedule new visit.	Checklist auto-assigned on visit create.	☐	
checklists-03	Matching	Division-specific rule	Create SERVICE-only template. Schedule SERVICE and INSTALL visits.	Checklist only on SERVICE visit.	☐	

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
checklists-04	Matching	Price book trigger rule	Link template to specific service item. Add that item to visit.	Checklist auto-syncs when line item added.	☐	
checklists-05	Matching	Exclude callbacks	Create template excluding callbacks. Flag visit as callback.	Template not applied to callback visit.	☐	
checklists-06	Completion	Mandatory for completion	Mark template mandatory. Try completing visit without finishing checklist.	Visit completion blocked.	☐	
checklists-07	Portal	Customer portal visibility	Complete checklist with portal visibility enabled.	Completed checklist visible in portal for that visit.	☐	
checklists-08	Templates	Duplicate template	Duplicate existing template.	Copy created as inactive with (copy) suffix.	☐	

11. Maintenance Plans

Plan templates, enrollment, billing, and scheduling.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
maintenance-01	Templates	Create plan template (full wizard)	Maintenance Plans → New template. Complete all wizard steps. Save.	Template active with visits, pricing, discounts, add-ons.	☐	
maintenance-02	Enrollment	Enroll customer in plan	Enroll via customer profile or visit. Select property, template, billing frequency.	DRAFT enrollment created.	☐	
maintenance-03	Activation	Accept and activate enrollment	Open enrollment. Click Accept & activate.	Status ACTIVE. Plan visits generated. Stripe subscription if configured.	☐	
maintenance-04	Scheduling	Schedule unscheduled plan visit	Dashboard → Unscheduled visits. Schedule with technician.	Visit created at due month. Linked to plan visit.	☐	
maintenance-05	Cancel	Cancel enrollment	Cancel active enrollment with reason.	Status CANCELLED. Stripe sub cancelled. Unscheduled visits skipped.	☐	
maintenance-06	Billing	Billing charge via API	POST /api/maintenance-plans/billing/{id}/charge for due period.	Stripe charge succeeds or stub-paid if no key.	☐	
maintenance-07	Dashboard	Dashboard metrics	View maintenance dashboard cards.	Enrollments, revenue, due billing, unscheduled counts load.	☐	
maintenance-08	Rachio	Rachio smart irrigation panel	View Smart irrigation card. Link Rachio in settings if needed.	Connection status shown. Links to settings.	☐	

12. Timesheets & Compensation

Shift tracking and pay preview.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
timesheets-01	Timesheets	Timesheets list — own entries	Log in as Tech. Open /timesheets. View own clock entries.	Entries show date, in/out, duration.	☐	
timesheets-02	Timesheets	Timesheets — manager view all	Log in as Manager. Filter by employee and date range.	All employee entries visible.	☐	
timesheets-03	Compensation	Pay period preview	As Manager, view pay preview section.	Per-employee: hours, OT, commission, projected payout.	☐	

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
timesheets-04	Settings	Compensation settings	Settings → Compensation. Set pay type defaults, OT threshold, commission basis. Save.	Settings persist. Affects pay preview calculations.	☐	

13. Marketing

Campaigns, social, SEO, ads, GBP, and referrals.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
marketing-01	Campaigns	Email blast campaign	Marketing → Campaigns → New. Setup → Audience (city filter) → Email content → Send now.	Campaign sends. Recipients table shows delivered/opened.	☐	
marketing-02	Campaigns	SMS blast campaign	Create SMS blast with message body. Send.	SMS delivered. STOP opt-out appended.	☐	
marketing-03	Campaigns	Drip campaign activation	Create DRIP with 3 steps. Activate.	Enrollment count shown. Steps process on schedule.	☐	
marketing-04	Campaigns	AI email generation	In email editor, enter prompt. Generate email.	HTML content generated. Editable in GrapesJS.	☐	
marketing-05	Social	Social post — submit for review	Social → Review queue. Compose post with caption and image. Submit for review.	Status PENDING_REVIEW. Appears in reviewer queue.	☐	
marketing-06	Social	Social post — approve and publish	As Admin/Manager, approve submission. Publish now.	Status PUBLISHED. Post appears on Meta.	☐	
marketing-07	Social	Social post — schedule	As Admin, schedule approved post for future datetime.	Status SCHEDULED. Auto-publishes at scheduled time.	☐	
marketing-08	Social	Social dashboard sync	Click Refresh on social dashboard.	Follower counts and recent posts update from Meta.	☐	
marketing-09	SEO	SEO — organic ranking map	Marketing → SEO. Select keyword. Refresh rankings.	Map shows positions by city. SerpAPI quota respected.	☐	
marketing-10	SEO	SEO — AI recommendations	Click Get recommendations.	3 tasks generated. Toggle complete. Delete task.	☐	
marketing-11	SEO	Website analytics dashboard	Select 7/28/30/90 day range. Refresh.	Page views, sessions, form submits, bookings metrics load.	☐	
marketing-12	SEO	Search Console connect	Connect Google Search Console OAuth. Select property.	Metrics: clicks, impressions, CTR, top queries load.	☐	
marketing-13	Ads	Google Ads dashboard	Marketing → Ads. Select date range. Refresh.	PPC campaigns show spend, impressions, clicks, conversions.	☐	
marketing-14	Ads	Meta Ads dashboard	View Meta ads tab with connected account.	Campaign metrics load.	☐	
marketing-15	GBP	GBP reviews summary	Marketing → Google Business. View review summary. Refresh.	Star breakdown, average rating, new last 7 days shown.	☐	
marketing-16	GBP	GBP reply to review	Reviews tab → Generate AI reply → Edit → Post reply.	Reply posted to Google. Review removed from unreplied list.	☐	
marketing-17	GBP	GBP create post	Posts tab → Generate post → Attach job photo → Publish.	Post published to Google Business Profile.	☐	
marketing-18	GBP	GBP upload photos	Photos tab → Select job photos → Upload to profile.	Photos appear on GBP.	☐	
marketing-19	Slack	Send GBP reviews to Slack	Click Send to Slack (Slack + channel configured).	Review cards posted to Slack channel.	☐	

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
marketing-20	Referrals	Referral program settings	Marketing → Referrals. Enable program. Set install/service rewards. Save.	Settings persist. Metrics grid loads.	☐	
marketing-21	Referrals	Referral pipeline and payout	Submit referral via public form. Convert to customer. Verify reward queue.	Pipeline shows submission. Reward queued. Retry works on FAILED.	☐	
marketing-22	Referrals	Stripe Connect for referrals	Click Verify Stripe Connect.	Connect onboarding completes. Payouts enabled.	☐	

14. Reporting

All dashboards and reports with correct data.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
reporting-01	KPI	KPI Dashboard — company overview	Open /reporting. Select YTD range. Verify 8 company metrics.	Service/install revenue, total, avg tickets, booking rate, conversion, active plans shown.	☐	
reporting-02	KPI	KPI Dashboard — technician cards	Verify solo tech cards: revenue, rev/hour, avg ticket, conversion, 5-star, callback rate.	Metrics match known test data for date range.	☐	
reporting-03	KPI	KPI Dashboard — CSR cards	Verify CSR booking rate, avg duration, booked revenue, plans sold.	Metrics calculated from call logs.	☐	
reporting-04	KPI	KPI Dashboard — crew cards	Verify install crew cards include callback rate.	Install crews show callback rate. Service crews do not.	☐	
reporting-05	KPI	KPI Dashboard — salespeople	Assign Sales role. Verify conversion, avg ticket, revenue sold.	Sales cards appear with correct metrics.	☐	
reporting-06	Insights	Business insights	Open /reporting/insights.	8 KPI cards: revenue MTD/YTD, pipeline, completion rate, etc.	☐	
reporting-07	Tech	Tech performance report	Open /reporting/tech-performance.	Per-tech table: visits, revenue, avg job size, hours.	☐	
reporting-08	Financial	Financial report	Open /reporting/financial.	Monthly invoiced vs collected for 12 months.	☐	
reporting-09	CSR	CSR report	Open /reporting/csr.	Call volume, book rate, disposition breakdown, per-agent table.	☐	
reporting-10	Estimates	Estimates report	Open /reporting/estimates.	Conversion rate and status breakdown.	☐	
reporting-11	Leads	Leads report	Open /reporting/leads.	Total leads, by status, by source.	☐	
reporting-12	Marketing	Marketing report	Open /reporting/marketing?days=30.	Website event counts by type and top pages.	☐	
reporting-13	Voice	Voice report	Open /reporting/voice.	Total calls, missed, avg duration (30 days).	☐	
reporting-14	Plans	Service plans report	Open /reporting/service-plans.	Enrollments, revenue, churn, due billing.	☐	
reporting-15	Invoices	Invoices AR aging	Open /reporting/invoices.	Aging buckets 0-30, 31-60, 61-90, 90+ with counts and balances.	☐	
reporting-16	Payments	Payments report	Open /reporting/payments.	Refund count and breakdown by payment method.	☐	

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
reporting-17	KPI	Date range control on KPI	Switch presets and custom range on KPI dashboard.	All sections refresh with new range data.	☐	

15. Customer Portal

Public self-service portal for customers.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
portal-01	Auth	Portal magic link login	Send portal link. Click link in email.	Customer logged in. Dashboard shows welcome and property.	☐	
portal-02	Visits	Portal — view visits	Navigate to Visits. View upcoming and past.	Visits listed with status, date, technician.	☐	
portal-03	Scheduling	Portal — schedule visit	Click Schedule a visit. Select property, reason, slot. Submit.	Visit created. Confirmation shown.	☐	
portal-04	Reschedule	Portal — reschedule visit	On scheduled visit, pick new slot. Reschedule.	Visit updated. Lead-time policy enforced.	☐	
portal-05	Cancel	Portal — cancel visit	Cancel scheduled visit. Confirm.	Status CANCELLED. Lead-time policy enforced.	☐	
portal-06	Invoices	Portal — view and pay invoice	Open Invoices. Click Pay on open invoice.	Stripe Checkout opens. Payment succeeds. Balance updates.	☐	
portal-07	Estimates	Portal — approve estimate	Open estimate link. Sign and approve.	Status APPROVED. Deposit flow if required.	☐	
portal-08	Maintenance	Portal — maintenance enrollments	View maintenance tab.	Active enrollments, upcoming visits, billing history shown.	☐	
portal-09	Rachio	Portal — Rachio zone run	On property with Rachio, run zone for 5 minutes (if allowed).	Zone run starts. Confirmation shown.	☐	
portal-10	Irrigation	Portal — irrigation guide	View published irrigation map and zone runtimes.	Map and programming guide display correctly.	☐	
portal-11	Offers	Portal — offers	View active offers. Click CTA button.	Offers display. External link opens.	☐	
portal-12	Referrals	Portal — submit referral	Referrals tab → Submit referral with friend details.	Submission recorded. History shows status.	☐	
portal-13	Feedback	Portal — feedback survey	Open /portal/{slug}/feedback/{token}. Rate 1-5 stars. Submit comment.	Survey saved. Thank you shown. Already-submitted on repeat.	☐	
portal-14	Settings	Portal feature gating	Disable portal features in settings. Verify hidden in portal nav.	Disabled sections not accessible.	☐	

16. Public Online Booking

Customer-facing booking widget at /book/{slug}.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
booking-01	Booking	Booking flow — happy path	Go to /book/{slug}. Enter contact, address with valid ZIP, pick slot. Book.	Confirmation shown. Visit created. Customer matched/created.	☐	
booking-02	Validation	Booking — invalid ZIP	Enter ZIP outside service areas.	Error: area not serviced. No slots shown.	☐	
booking-03	Validation	Booking — slot race condition	Select slot, delay submit until slot taken.	409 error. User prompted to pick another slot.	☐	

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
booking-04	Validation	Booking — Do Not Service block	Book with phone/email of DNS customer.	403 blocked. No visit created.	☐	
booking-05	Settings	Booking disabled	Disable online booking in settings. Visit /book/{slug}.	404 or not available message.	☐	
booking-06	Notifications	Booking notification	Complete booking. Verify visit-scheduled notification sent.	Customer receives SMS/email per communication settings.	☐	

17. Payments

Invoice payment and card-on-file flows.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
payments-01	Invoice pay	Public invoice payment	Open /pay/{token}. Click Pay. Complete Stripe Checkout.	Payment succeeds. Invoice balance updates. Receipt sent if configured.	☐	
payments-02	Invoice pay	Already-paid invoice	Open pay link for fully paid invoice.	Thank you / already paid message. No pay button.	☐	
payments-03	Invoice pay	Refunded invoice	Open pay link for refunded invoice.	Not payable message shown.	☐	
payments-04	Visit pay	Visit checkout payment	From visit detail, Collect payment. Complete Stripe.	Visit marked Paid. Invoice synced.	☐	
payments-05	Deposit	Estimate deposit payment	Approve estimate with deposit via portal. Pay deposit.	Deposit recorded. Install flow triggered if configured.	☐	
payments-06	Card on file	Card setup success/cancel	Use secure card-add link. Complete or cancel Stripe setup.	Success page at /pay/card/success. Cancel at /pay/card/cancelled.	☐	
payments-07	Refunds	Issue refund	As Admin, refund payment on invoice from /customers/invoices.	Refund processed in Stripe. Invoice status updated.	☐	

18. Settings (Global & Feature Configuration)

All company configuration before go-live.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
settings-01	Company	Company profile	Settings → Company. Edit name, address, timezone, support email, phone. Save.	Profile persists. Timezone affects scheduling.	☐	
settings-02	Company	Email branding	Upload email logo. Set sender display name.	Logo appears in outbound emails and portal header.	☐	
settings-03	Company	Business hours (company)	Set Mon–Fri open hours. Save.	Hours persist. Used for voice after-hours routing.	☐	
settings-04	Appearance	Appearance theme	Settings → Appearance. Toggle Light/Dark/System.	Theme applies immediately.	☐	
settings-05	Communications	Communications templates	Settings → Communications. Edit visit-scheduled SMS template. Insert merge field. Save.	Template saves. Test send delivers with merged data.	☐	
settings-06	Communications	Review link tracking	Enable review request notification. Send test. Click review link.	Sent/clicked stats increment in tracking panel.	☐	
settings-07	Employees	Create employee with role	Settings → Employees → Create. Set role TECH, division, service areas, compensation. Save.	Employee active. Can log in. Appears on schedule filters.	☐	

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
settings-08	Employees	Archive and restore employee	Archive employee. Verify hidden from active. Restore.	Status toggles. Login blocked while archived.	☐	
settings-09	Employees	Set employee password (Admin)	As Admin, set password for employee.	Employee can log in with new password.	☐	
settings-10	Crews	Create and manage crew	Create install crew with division INSTALL. Add members.	Crew appears on schedule. KPI crew card shows callback rate.	☐	
settings-11	Service areas	Service areas and ZIPs	Create area. Add ZIP codes. Use in scheduling and booking.	ZIPs validate booking and quick-add.	☐	
settings-12	Parts	Parts suppliers	Search Google Places. Add supplier. Activate.	Supplier appears in Parts run on visits.	☐	
settings-13	Booking	Booking settings	Enable booking. Set slug and lead time. Save.	Public /book/{slug} accessible.	☐	
settings-14	Portal	Customer portal settings	Enable portal. Toggle feature flags (visits, estimates, invoices, etc.). Set reschedule/cancel lead times.	Portal reflects toggles. Policies enforced.	☐	
settings-15	Portal	Portal offers CRUD	Create offer with title, description, URL. Toggle active. Delete.	Active offers appear in portal.	☐	
settings-16	Leads	Lead settings	Add lead source tags. Set default assignee. Enable notify on website lead.	Sources available on lead forms. Notifications fire.	☐	
settings-17	Inbox	Inbox settings	Set Twilio phone, from email. Send test email. Register webhooks.	SMS and email outbound work. Webhooks registered.	☐	
settings-18	Estimates	Estimate settings	Set expiry days, deposit defaults, install duration, supplier email, auto-email parts.	New estimates inherit defaults.	☐	
settings-19	Invoices	Invoice settings	Set prefix, payment terms, footer.	New invoices use settings.	☐	
settings-20	Intake	Customer intake required fields	Mark phone and address required. Create customer without them.	Validation blocks incomplete create.	☐	
settings-21	Integrations	Integrations — API keys	Generate WEBSITE API key. Copy raw key. Revoke.	Key works for spoke integration. Revoked key fails.	☐	
settings-22	GBP	Google Business Profile connect	Settings → Integrations → GBP. OAuth connect. Select location.	Connected. Marketing GBP pages load data.	☐	
settings-23	Google Ads	Google Ads connect	OAuth connect. Enter developer token. Select customer account.	Ads dashboard loads campaign data.	☐	
settings-24	Meta Ads	Meta Ads connect	Enter token. Select ad account.	Meta ads tab loads data.	☐	
settings-25	Slack	Slack GBP reviews	Configure Slack channel. Send test card.	Test card appears in Slack.	☐	
settings-26	SEO	SERP rankings settings	Add GBP and organic keywords. Add target cities. Save.	SEO ranking maps use configured keywords/cities.	☐	
settings-27	Rachio	Rachio API key	Settings → Maintenance. Enter Rachio API key. Test connection.	Connection succeeds. Controllers listed.	☐	

19. Data Migration (Housecall Pro)

One-time import from Housecall Pro. Run in staging first.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
migration-01	Migration	HCP connect and preview	Settings → Data migration. Enter HCP API key. Preview import counts.	Preview shows expected record counts per step.	☐	
migration-02	Migration	Run migration batches	Start migration. Run batches for tags, zones, employees, materials, services, customers, jobs.	Each step completes. Counts increment. Auto-continue works.	☐	
migration-03	Migration	Skip and focus steps	Skip a step. Focus on specific step. Reset step.	Step state changes correctly.	☐	
migration-04	Migration	Pause and resume	Pause migration mid-batch. Resume.	State preserved. Continues from pause point.	☐	
migration-05	Migration	Rollback (destructive)	In staging only: run rollback with confirmation.	All HCP-imported records deleted.	☐	

20. Cross-Integration End-to-End Scenarios

Full workflows spanning multiple modules. Run these last.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
integrations-01	E2E	Lead → customer → visit → complete → invoice → pay	Create lead. Convert. Schedule visit. Complete with checklist. Generate invoice. Customer pays via portal.	End-to-end data flows correctly. Balances zero.	☐	
integrations-02	E2E	Inbound call → book → dispatch → complete	Receive call. Book appointment. Tech goes On my way. Completes visit.	Call disposition BOOKED. Visit completed. KPIs update.	☐	
integrations-03	E2E	Online booking → notification → schedule board	Customer books online. Verify notification. Find on schedule board.	Visit on board. Customer and property created.	☐	
integrations-04	E2E	Estimate → send → portal approve → deposit → schedule install	Create estimate. Send. Customer approves in portal. Pays deposit. Appears in needs-scheduling.	Full sales-to-install pipeline works.	☐	
integrations-05	E2E	Maintenance plan enroll → activate → schedule visits → bill	Enroll customer. Activate. Schedule plan visits. Verify billing period.	Plan lifecycle complete.	☐	
integrations-06	E2E	Marketing campaign → track opens → reporting	Send email campaign. Open email (click link). Check reporting/marketing events.	Open tracked. Events appear in marketing report.	☐	
integrations-07	E2E	GBP review → Slack card → reply in CRM	Receive new Google review. Send to Slack. Reply from Marketing GBP page.	Slack card posted. Reply on Google.	☐	
integrations-08	E2E	Referral submit → convert → payout	Customer submits referral. Referred becomes customer with paid visit. Verify reward payout.	Reward queued and transferred via Stripe Connect.	☐	
integrations-09	E2E	Do Not Service enforcement everywhere	Flag customer DNS. Attempt: book online, New visit, schedule plan visit, send notifications.	All paths blocked. No notifications sent.	☐	
integrations-10	E2E	Role-based access sweep	Log in as each role. Verify allowed modules accessible. Verify forbidden actions blocked.	TECH cannot delete customers. INSTALLER cannot manage enrollments. CSR can schedule.	☐	

21. Pre-Go-Live Checklist

Final verification before handing the CRM to your team for daily use.

ID	AREA	TEST CASE	STEPS	EXPECTED RESULT	PASS	NOTES
golive-01	Env	Production environment variables	Verify all required env vars set: DATABASE_URL, NEXTAUTH_SECRET, STRIPE keys, TWILIO, SENDGRID, GOOGLE_OAUTH, SLACK_BOT_TOKEN, BLOB, OPENAI (optional).	No 503 errors on integration pages. Health checks pass.	☐	
golive-02	Twilio	Twilio webhooks registered	Settings → Inbox → Register webhooks. Verify in Twilio console.	Inbound SMS and voice route to CRM.	☐	
golive-03	Stripe	Stripe webhooks configured	Verify Stripe webhook endpoint points to /api/stripe/webhook.	Payment confirmations process automatically.	☐	
golive-04	Google	Google OAuth redirect URIs	Verify all callback URIs registered in Google Cloud Console.	GBP, Ads, Search Console OAuth flows complete without error.	☐	
golive-05	Meta	Meta webhooks verified	Settings → Meta webhooks. Verify webhook in Meta developer console.	Social DMs and post insights flow in.	☐	
golive-06	Areas	Service areas cover all ZIPs	Review service area ZIP lists match your actual service territory.	Booking and scheduling accept valid ZIPs only.	☐	
golive-07	Price book	Price book populated	Import or manually enter all services and materials used in the field.	Techs can add line items from complete catalog.	☐	
golive-08	Checklists	Checklists configured	Create and activate checklists for SERVICE and INSTALL divisions.	Visits auto-receive correct checklists.	☐	
golive-09	Communications	Communication templates reviewed	Review all 12 notification templates. Send test for each channel.	Messages professional and merge fields correct.	☐	
golive-10	Employees	Employee accounts created	Create accounts for all team members with correct roles.	Each person can log in and sees appropriate modules.	☐	
golive-11	Portal	Customer portal tested	Send portal link to test customer. Walk through all enabled features.	Portal works on mobile and desktop.	☐	
golive-12	Booking	Online booking tested	Complete test booking from public URL on mobile.	Booking works end-to-end.	☐	
golive-13	Ops	Backup and rollback plan	Document database backup procedure. Test restore in staging.	Can recover from bad data entry or migration mistake.	☐	

End of Storm Sprinklers CRM Pre-Launch Test Plan

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